PRITI MAHAJAN

Data Scientist | Data Analyst | Machine Learning Engineer

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[LinkedIn](#) | [GitHub](#) | [Live Portfolio](#)

PROFESSIONAL SUMMARY

Analytical and detail-oriented **Data Scientist & Data Analyst** with a strong foundation in statistical modeling, predictive analytics, and machine learning pipeline development. Proficient in transforming raw structural and unstructured data into actionable business insights using **Python, SQL, and Power BI**. Demonstrates proven capability in building end-to-end Machine Learning systems from scratch, paired with professional frontend cross-functional engineering experience (**React.js, Next.js**) to deploy models as interactive, high-fidelity user web applications. Highly adept at algorithmic problem-solving and cross-team collaboration within agile enterprise workflows.

TECHNICAL SKILLS

- **Core Programming:** Python (Pandas, NumPy), Advanced SQL (MySQL), Java, JavaScript (ES6+)
- **Machine Learning & AI:** Scikit-learn, Regression Analysis, Statistical Classification, Recommendation Systems, Feature Engineering, Hyperparameter Tuning, Model Evaluation Metrics
- **Natural Language Processing (NLP):** Text Preprocessing, TF-IDF Vectorization, Tokenization
- **Data Visualization & BI:** Power BI (DAX, Interactive Dashboards), Excel (VLOOKUP, Pivot Tables), Matplotlib, Seaborn
- **Secondary Frontend Skills:** React.js, Next.js, Tailwind CSS, HTML5, CSS3
- **Backend & Tools:** Spring Boot, RESTful Web APIs, Git, GitHub, VS Code, Jupyter Notebook, Google Colab

PROFESSIONAL EXPERIENCE

Frontend Developer Intern | Reet Technology
Pune, India

- Developed responsive web applications using **React.js, Next.js, and Tailwind CSS**.
- Built reusable UI components and integrated **REST APIs** for dynamic applications.

- Used **Git & GitHub** for version control, collaboration, and code management.
- Performed testing, debugging, and performance optimization to ensure smooth user experience.
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🚀 TECHNICAL PROJECTS

1. Online Course Recommendation System

- **Business Problem & Objective:** E-learning platforms struggle with user churn due to generic item discovery. The objective was to build an automated recommendation engine to drive personalized content engagement.
- **Methodology & ML:** Conducted exploratory data analysis (EDA) and robust feature processing utilizing **Pandas** and **NumPy**. Built a personalized content-filtering recommendation pipeline leveraging **Scikit-learn** to match user historical consumption patterns with course feature vectors.
- **Technologies:** Python, Pandas, NumPy, Scikit-learn, Jupyter Notebook
- **Business Impact:** Improved recommendation relevance metrics, significantly lowering discovery time and unlocking higher user engagement simulation indicators.
- **Codebase:** github.com/PritiHM

2. CO_2 Emissions Prediction & Regression Pipeline

- **Objective:** Designed a predictive analytics model to accurately forecast industrial vehicle carbon emissions based on manufacturing and engine performance metrics.
- **Data Engineering:** Conducted exhaustive data cleaning, handled missing attributes, executed **One-Hot Encoding** for categorical properties, and streamlined workflows via an automated **Column Transformer** pipeline.
- **Model Assessment:** Built and compared multiple models including *Linear Regression*, *Decision Tree Regression*, and *Extra Trees Regression*. Evaluated models based on Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R^2Score.
- **Results:** The **Extra Trees Regression** model outperformed all other baselines, securing an optimal R^2Score and demonstrating exceptional generalization limits against unseen test splits.

- **Codebase:** github.com/PritiHM

3. Automated Spam Email Classification System (NLP)

- **Objective:** Developed an end-to-end Text Classification pipeline designed to automatically filter out spam emails from legitimate enterprise communications.
- **NLP Pipeline:** Applied Natural Language Processing techniques for text preprocessing, including custom punctuation removal and stop-word filtering. Transformed raw corpus strings into high-density numeric arrays using **TF-IDF Vectorization**.

- **Machine Learning Classifier:** Trained and optimized a Scikit-learn Classification model to distinguish text features with high precision, lowering false-positive flags on critical operational emails.

- **Codebase:** github.com/PritiHM

4. Premium Personal Portfolio & Model Showroom

- **Objective:** Engineered a modern, single-page professional web ecosystem to showcase data science models and deployment endpoints.
- **Architecture:** Developed using **React.js** and **Tailwind CSS**, highlighting production-ready responsive layout configurations and smooth animation assets.

- **Live Application:** priti-portfolio-eight.vercel.app

🎓 EDUCATION

- **Bachelor of Science in Computer Science (B.Sc. CS)**

  North Maharashtra University | **Year of Passing: 2022**
- **Higher Secondary Certificate (HSC)**

  Maharashtra State Board | **Year of Passing: 2019**
- **Secondary School Certificate (SSC)**

Maharashtra State Board | **Year of Passing: 2017**

🏆 PROFESSIONAL CERTIFICATIONS

- **ExcelR Data Science Professional Certification** (Comprehensive training in statistical modeling, machine learning algorithms, and analytics)
- **NASSCOM Industry-Aligned Certification**
- **Professional Backend Developer Certification**

💡 SOFT SKILLS & CORE COMPETEN/CIES

- Analytical Thinking & Core Problem Solving
- Critical Evaluation & Data-driven Decision Making
- Articulate Professional Communication & Technical Documentation
- Agile Team Collaboration & Cross-functional Adaptability
- Project Lifecycle Management & Time Efficiency

🌐 LANGUAGES

- **English** (Professional Proficiency)
- **Hindi** (Professional Proficiency)
- **Marathi** (Native/Bilingual Proficiency)